Dataset and Benchmark for Ship Detection in Complex Optical Remote Sensing Image

Jianming Hu, Xiyang Zhi, Tianjun Shi, Junjie Wang, Yuelong Li, and Xiaogang Sun

***Abstract*— Ship detection plays a pivotal role in numerous military and civil applications, yet detecting ships in complex maritime and aerial environments remains a challenging task. While several publicly available datasets for ship detection have been introduced by researchers, most of them do not adequately address the impacts of diverse and intricate environmental factors, which makes the trained algorithms difficult to apply for practical application scenes involving clouds, sea clutter, complex lighting, and facility interferences, limiting the effectiveness and robustness of the detection models. To advance the field of ship detection method research, we propose a high-quality dataset named ship collection in complex optical scene (SCCOS), which is obtained from multiple platform sources including Google Earth, Microsoft map, Worldview-3, Pleiades, Orbview-3, Jilin-1, and Ikonos satellites. The dataset comprehensively considers complex scenes such as thin clouds, mist, thick clouds, light shadows, sea clutter, and port facilities. Additionally, we conduct experiments on this dataset with 11 representative detection algorithms and establish a performance benchmark, which can provide the theoretical basis and practical reference for the design and optimization of subsequent ship detection models. The latest dataset is available at: https://github.com/JimmyRSlab/Dataset-and-Benchmark-for-Ship-Detection-in-Complex-Optical-Remote-Sensing-Image.**

***Index Terms*— Complex scene, detection benchmark, environmental interferences, optical remote sensing image, ship detection.**

I. Introduction

SHIPS are the primary carrier of maritime activities, and the intelligent detection of ship targets on the sea surface through optical remote sensing images has garnered global research interest [1]. The core task of ship detection based on optical images involves identifying the presence of ships, accurately classifying and pinpointing their locations. It is of great significance for applications including fisheries management, maritime rescue, port traffic maintenance, and sea area monitoring. The accuracy of detection technology directly determines the safety and timeliness of military and civilian applications. Nonetheless, due to the variability of the target's scale and direction [2], as well as the interference of various complex environmental factors, ship detection in complex scenes remains a challenging task.

Regarding the issue of ship detection, researchers have proposed multiple publicly available datasets, mainly including multiscale ship images of typical sea and port scenes [3]. However, due to the long-range and wide coverage characteristics of remote sensing imaging, optical detection is easily affected by various factors such as sea surface conditions, weather conditions, lighting conditions, and imaging detector parameters. This is reflected in the presence of natural environmental factors such as mist, sea clutter, and clouds in the image, which directly affect the performance of ship target detection in the image. The existing datasets mainly focus on the characteristics of the target such as type and scale, and there are few datasets constructed from the perspective of interference factors in the real environment, which to some extent limits the robustness of ship detection algorithms.

Specifically, the current ship detection datasets face the following bottleneck issues.

1) Images from different platforms exhibit distinct degradation characteristics, and existing ship detection datasets predominantly stem from a singular platform, resulting in inadequate coverage of ship target characteristics under diverse platforms and environmental conditions.
2) Previous public ship detection datasets lack comprehensive coverage of various natural environmental interferences, such as thin clouds, light mist, thick clouds, lighting shadows, sea surface clutter, and port facilities. This limitation hinders the adaptability of trained algorithms to real-world complex application scenes.

This article constructs a complex optical scenes dataset for ship detection based on Google Earth, Microsoft map, Worldview-3, Pleiades, Ikonos, Orbview-3, and Jilin-1 satellite images, named ship collection in complex optical scene (SCCOS). 4639 images are carefully selected and 19 078 instances are annotated using the oriented bounding box (OBB) format. Moreover, this dataset covers various natural environmental factors including thin clouds, mist, thick clouds, lighting and shadows, sea clutter, and port facilities, which can support research and training optimization of ship target detection models. In addition, based on the established dataset, this article also utilizes 11 representative detection methods in recent years to fairly evaluate the dataset, providing a benchmark for subsequent algorithm application. The contributions of this work can be summarized as follows.

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TABLE I
DEVELOPMENT REVIEW OF SHIP DETECTION DATASETS

Dataset	Year	Organizations or competitions	Data sources	Ship instance number
NWPU VHR-10	2014	Northwestern Polytechnical University	Google Earth, Vaihingen	302
HRSC2016	2016	Northwestern Polytechnical University	Google Earth	2976
LEVIR	2017	Beihang University	Google Earth	3025
DOTA	2017	Wuhan University	Google Earth, Jilin-1, Gaofen-2	43736
DIOR	2018	Northwestern Polytechnical University	Google Earth	About 63000
TGRS-HRRSD	2018	University of Chinese Academy of Sciences	Google Earth, Baidu Map	3886
Airbus Ship Detection	2018	Kaggle	SPOT	81723
Ships in Satellite Imagery	2018	Kaggle	PlanetScope	1000
MASATI-v2	2019	University of Alicante	Microsoft Bing Map	4157
FGSD	2020	Beijing University of Posts and Telecommunications	Google Earth	5634
PSDS	2020	Universidad Nacional de San Agustín de Arequipa	PERUSAT-1	9662
MSDS	2020	Universidad Nacional de San Agustín de Arequipa	Google Earth	4710
ShipRSImageNet	2020	Tsinghua University	xView, HRSC2016, Airbus Ship, Gaofen-2, Jilin-1	17573
FGSCR-42	2021	Beihang University	Google Earth	9320
S2-SHIPS	2021	IMT Atlantique	Sentinel-2	1053
GF1-LRSD	2021	Chinese Academy of Sciences	Ganfen-1	7172
VHRShips	2022	Istanbul Technical University	Google Earth	11179
Ours	2023	Harbin Institute of Technology	Google Earth, Microsoft map, Worldview-3, Pleiades, Ikonos, Orbview-3 and Jilin-1	19078

1) The proposed ship detection dataset comprises slices of multiple satellite sources from over 100 typical military and civilian port scenes worldwide, taking into account the diversity of port scenes, platform sources, and military and civilian target types.
2) The proposed ship detection dataset comprehensively considers various natural environmental interferences, including thin clouds, mist, thick clouds, shadows, sea clutter, and port facilities, which enables the adequate model training and performance evaluation for ship detection algorithms in complex scenes.
3) A ship detection benchmark is presented based on the constructed dataset and 11 representative detection methods are employed for training and assessment, which contributes to catalyze the development and deployment of advanced ship detection technologies.

The subsequent sections of this article are structured as follows. Section II presents an overview of development process about ship detection datasets and the corresponding detection methodologies. In Section III, we illustrate the production process of the dataset and its main image characteristics. Section IV offers a comparative analysis by using 11 representative detectors on the proposed dataset. Finally, we conclude and summarize in Section V.

II. RELATED WORKS

A. Ship Detection Datasets

Deep learning methods play a pivotal role in the field of computer vision, providing a powerful tool for computer vision tasks. Their exceptional image understanding and data processing capabilities have significantly advanced the development of remote sensing image target detection technology. By mining the limited feature set of training set images, the deep learning models can infer the high-quality multilevel semantic features of ship targets, enabling the effective detection of ships in complex application scenes. They are extensively applied in various military and civilian applications, including maritime traffic management, military target monitoring, and marine environmental protection. Considering that the richness and diversity of training data directly influence the performance of deep learning networks, numerous researchers have successfully released multiple high-quality ship detection datasets (as seen in Table I) to enhance the detection performance for multiscale objects under complex scene conditions.

In 2014, Northwestern Polytechnical University (NWPU) released the NWPU VHR-10 dataset [4], which consists of images extracted from Google Earth and Vaihingen sources. Moreover, it comprises ten categories of remote sensing objects such as ships and aircraft, with a total of 302 ship instances. In 2016, the HRSC2016 dataset was released by NWPU [5], whose image scenes were extracted from six famous ports in Google Earth such as Norfolk Naval Base, San Diego Naval Base and Murmansk harbor. The dataset contains 1068 images in total, while only 1061 images are annotated, resulting in a total of 2976 ship instances. In 2017, the LEVIR dataset [6] was provided by Beihang University, with data sourced from Google Earth. It encompasses three object categories: aircraft, ship, and fuel tanks, including 3025 ship instances. The DOTA dataset [7], released by Wuhan University in 2017, were extracted from Google Earth, Jilin-1, and Gaofen-2 satellite images. It includes 15 different categories, and a total of 43 736 instances of ships.

In 2018, the DIOR dataset was launched by NWPU. The image data consisted of 23 463 remote sensing images collected from Google Earth and 190 288 object instances, covering 20 categories and about 63 000 ship instances. The spatial resolution of the dataset ranged from 0.5 to 30 m. In 2018, the TGRS-HRRSD dataset [8] was provided by the Chinese Academy of Sciences, with data sourced from Google Earth and Baidu Map, containing 3886 ship instances. Furthermore, Airbus released the Airbus Ship Detection dataset

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on the Kaggle platform to seek faster and more accurate ship detection solutions. The dataset consisted of 192 556 images and 81 723 ship instances. Despite the large number of ship instances, the dataset mainly includes ship image scenes with good lighting conditions and less interference from scene elements. The ships in Satellite Imagery dataset was also released on the Kaggle platform in 2018, with data extracted from the planet satellite images of San Francisco Bay and San Pedro Bay, containing 4000 images. In this dataset, 1000 images are labeled as the ship category. The ship images were centered on the single ship, containing ship instances with different sizes, orientations, and atmospheric conditions.

In 2019, the MASATI-v2 dataset [9] was released by the University of Alicante. The data were derived from Microsoft Bing Map and included 7389 images, containing 4157 instances of small-scale ships. In 2020, the FGSD dataset [10] was introduced by Beijing University of Posts and Telecommunications, and the images of the dataset were sourced from Google Earth. 5634 ship instances were collected from 17 large ports in four countries in total, and 43 categories were specifically labeled. In 2020, PSDS and MSDS dataset [11] were released by Universidad Nacional de San Agustín de Arequipa. Researchers built the above two datasets based on different satellite images. Specifically, the PSDS data were extracted from 22 RSI images taken by the PERUSAT-1 satellite, containing 9662 ship instances and 1310 images. The MSDS data were extracted from Google Earth and 2993 images, containing 4710 ship instances. Also, the ShipRSImageNet dataset [12] was published by Tsinghua University, which contained 3435 images and 17 573 ship instances. The data were collected from a variety of platforms and datasets, including: 532 images extracted and resized from the xView dataset with 930 × 930 pixels, 1057 images sourced from HRSC2016 dataset, 1846 images extracted from FGSD dataset, 21 images from Airbus ship dataset, and 17 images extracted from the Gaofen-2 and Jilin-1 satellite data.

In 2021, Beihang University released the FGSCR-42 dataset [13] to address the Fine-grained visual categorization. This dataset was sourced from Google Earth and several popular remote sensing datasets such as DOTA, HRSC2016, and NWPU VHR-10. The whole dataset included 42 categories and 9320 images, with each image corresponding to an individual ship instance. In 2021, the S2-SHIPS dataset [14] was published by IMT Atlantique. The data were derived from Sentinel-2, and the images were annotated with 1053 ship instances at the pixel level. Also, the GF1-LRSD dataset [15] was introduced by the Chinese Academy of Sciences, with data obtained from the Gaofen-1 satellite, containing 7172 instances of micro-sized ships. In 2022, the VHRShips dataset [16] was released by the Istanbul Technical University, with images extracted from the Google Earth. It included 5312 images containing ships and 1000 images without ship targets, totaling 11 179 ship instances.

From the above descriptions, it can be inferred that the current ship detection datasets suffer from a limited image source, primarily relying on Google Earth, which poses challenges in ensuring image diversity. To address this issue, we propose the SCCOS dataset, comprising images from over 100 typical ports in various countries around the world, sourced from multiple satellite platforms including Google Earth, Microsoft map, Worldview-3, Pleiades, Ikonos, Orbview-3, and Jilin-1. This diversification enhances the reliability, authenticity, and scene variety of the dataset.

B. Ship Detection Methods

Traditional ship detection methods are mainly based on shape and texture features to locate and classify targets [17], [18]. However, these methods are susceptible to the influence of factors such as complex environmental elements and target scale, which limits the robustness of the algorithm. In recent years, with the continuous development of deep learning models, learning-based object detection methods have possessed the advantages of self-adaptability, robustness, powerful feature representation ability, and end-to-end training [19], showing a trend toward gradually replacing traditional machine-learning methods in the field of ship detection [20].

Learning-based ship detection algorithms are mainly divided into two categories: two-stage and one-stage detection. The two-stage algorithms divide the detection process into two steps: first generating a series of proposed regions that may contain the target, and then utilizing detection heads to perform bounding box regression and target classification tasks. The typical representatives of two-stage detection algorithms are the R-CNN series [21]. In contrast, the one-stage algorithms skip the step of proposing candidate regions, simplifying the detection problem to a regression problem. The typical one-stage object detection algorithms mainly include YOLO [22] and SSD [23].

In previous research, scholars adapted the general object detection algorithm to the ship target detection task by modification [24]. In 2016, a method based on the hierarchical selective filtering (HSF) was introduced [25], which optimizes the multiscale feature extraction strategy based on the CNN architecture, showing excellent ship target detection performance. Unlike general object detection, ship targets have the characteristics of large-scale variation, dense arrangement, large aspect ratio and random direction. Applying horizontal bounding boxes to describe targets leads to the problem of excessively small target area proportion. Therefore, researchers proposed the use of rotated bounding boxes for more accurate position and orientation distribution of targets [26], leading to the development of various ship detection models.

In 2017, R2CNN [27] was improved upon Faster R-CNN [28] by introducing the concept of rotated bounding boxes. While the region proposal network (RPN) network still output horizontal proposal regions, it represented the rotating bounding box using the clockwise coordinates of the first two points and the height of the bounding box $(x_1, y_1, x_2, y_2, w, h)$. In 2018, RRPN network was presented [29], which introduced six different directions of rotated angles into the horizontal box of Faster R-CNN, thus generating oriented proposal regions. These regions were then converted into the same size through

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TABLE II
IMAGE SOURCE STATISTICS OF THE PROPOSED DATASET

Data sources	Spatial resolution	Image type	Spectral range	Image number	Average pixel size	Slice Numbers
Worldview-3	0.31 m	Panchromatic	450-800 nm	21	(34363, 31996)	129
Jilin-1	0.72 m	Panchromatic	500-800 nm	31	(16299, 13392)	119
Ikonos	1.00 m	Panchromatic	450-900 nm	2	(16276, 50524)	12
Orbview-3	1.00 m	Panchromatic	450-900 nm	4	(8016, 24159)	9
Pleiades	0.5 m	Panchromatic	480-820 nm	7	(15202, 13554)	17
Google Earth	0.42-1.02 m	RGB	R: 620-780 nm G: 500-560 nm B: 430-470 nm	418	(20224, 14592)	1989
Microsoft map	0.40-2 m	RGB		454	(17408, 11520)	2364

the RRoI pooling, achieve the accurate prediction of the target in the original input image.

Subsequently, RoI Transformer [30] proposed the RoI leaner module to supervise the rotation, which can convert horizontal RoI into RRoI and extract rotation-invariant features, enabling effective detection of rotated targets. Although the above two-stage detection network were effective, their detection speed were limited by the complex network structure. In response to this situation, in 2021, Lin et al. [31] proposed the single-stage network R3Det based on RetinaNet. It adopted a coarse-to-fine strategy to reencode the position information of predicted boxes to the corresponding feature points, achieving feature map reconstruction and feature alignment. In 2021, a similar single-stage detector named S2A-Net was proposed [32], which consisted of a feature alignment module (FAM) and an orientation detection module (ODM). The FAM module adaptively aligned features using alignment convolution (AlignConv) and the ODM module encoded angle information using active rotating filters (ARFs) to produce orientation-sensitive features.

In 2023, the CMDet network [33] utilized 1-D Fourier contour coefficients sampled from clockwise contours at equal angle intervals to represent the rotated bounding contour information. It then predicted the position of the center point, enabling precise detection of anchor-free, arbitrarily oriented ship targets. Furthermore, the FFESSD network [34] introduced the feature fusion module (FFM) to combine features from each layer in the feature pyramid, improving the accuracy of small target detection. With the research progress of Transformers in the field of remote sensing image processing, researchers attempted to incorporate the vision transformer into ship target detection [35]. They achieved an adaptive learning mechanism for rotated object detection by mining local and long-range dependencies, extracting rich property information of interested targets from complex scenes. A concise and effective one-stage anchor-free contour modeling detector called CMDet is proposed for accurate ship detection [33], which is combined with Fourier contour modeling strategy and inherent geometry loss. Also, Zhang et al. [36] designed a remote sensing-image synthesis pipeline (RS-ISP) to address the lack of on-orbit remote sensing images for ship detection. Moreover, Guo et al. [37] proposed a shape-aware feature learning method that contains shape-aware RoI extractor (SRE) and shape-aware rotated RoI align (SRRA) to mitigate the misalignments in ship feature extraction.

The aforementioned detection algorithms belong to the field of deep learning, and are principally grounded in statistical analysis, requiring extensive data training. Current ship detection datasets seldom incorporate samples that comprehensively represent various environmental factors, constraining the effectiveness and robustness of algorithms. Consequently, our dataset meticulously accounts for the intricacies of remote sensing image scene, encompassing a wide range of environmental disturbances including thin clouds, mist, thick clouds, shadows, sea clutter, and port facilities. This breadth of coverage is instrumental in enhancing the refinement of ship target detection methodologies.

III. DATASET CONSTRUCTION

A. Construction Method

In order to achieve data diversity, we obtain image slices of port scenes from Google Earth, Microsoft map, and satellites images including Worldview-3, Jilin-1, and Pleiades. Table II provides the resolution, quantity, size information of the source images of each satellite, as well as the corresponding number of generated slice images. In addition, to ensure the richness and variety of the scene, we select typical ports from countries around the world such as China, Russia, USA, Japan, Singapore, and The Netherlands. Therefore, the final image scene includes more than 100 large and medium-sized military and civilian ports, including Norfolk Port and San Diego Port in USA, Yokosuka Port in Japan, Shanghai Port and Zhoushan Port in China, Singapore Port, and Busan Port in South Korea. It is worth noting that since we collect platform data from different sources as much as possible, most of the images are RGB images, and a few satellite images obtained are panchromatic images. Therefore, the final data include 4353 RGB images and 286 grayscale images.

The production of the dataset mainly includes the following process.

1) To improve the accuracy of annotation, manual filtering and position confirmation are performed on large-scale satellite images, followed by uniform cropping to obtain 1024×1024 pixel scenes featuring ship targets.
2) We apply LabelImg2 software to annotate all sliced images. Considering that rotational box annotation provides a more precise representation of the position and orientation of ships compared to horizontal box annotation, rotational box annotation is employed uniformly. For ease of use, we have formatted the annotations in the widely recognized PASCAL VOC format

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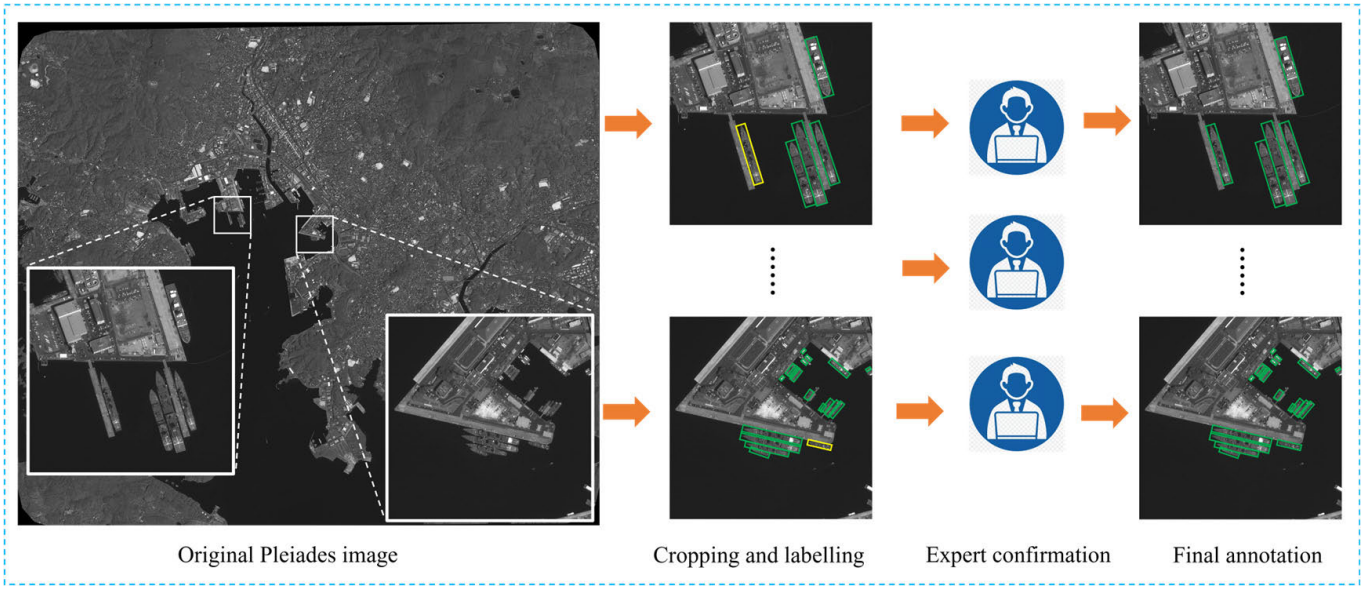


Fig. 1. Production process of the dataset images.

$(x_{p1}, y_{p1}, w, h, \text{angle})$, which accurately describes the position and posture of the target object using the center point coordinates (x, y), width (w), height (h), and rotation angle (angle) of the rotating rectangle.

3) It is important to note that due to the random distribution of ships, some image slices may only include partial views of ships. Addressing this, combined with the detection capabilities of current mainstream algorithms, we decide to label targets in the slice only if the area of incomplete ships exceeds half of the complete area of the ship in that scene (which can be derived from large-scale satellite images). To enhance the accuracy of this process, multiple interpretation experts are invited to evaluate and annotate, ultimately determining the precise number of incomplete ships that required annotation. As shown in Fig. 1, taking the Pleiades image input as an example, we provide the production process of dataset images.

It is worth putting out, due to the complexity of the scene, conditions such as clouds, mist, sea clutter, and lighting may appear single or randomly combined in each image. Explaining each category one by one may appear cumbersome and not visually appealing. We use the "difficult" flag in the VOC format to represent the complexity of the image scene. Taking into account factors such as target size, lighting changes, background clutter, and image quality, we believe that the "difficult" flag of images that are prone to false alarms and missed detections for classical detection algorithms is set to 1. In addition, due to the scarcity of cloud and mist scenes, we utilize the cloud and mist inversion technique to simulate and generate partial images, thus enriching the diversity of the image scenes.

Finally, we have generated 4639 annotated scene slices from 937 large-scale satellite images, including 19 078 ship instances. Fig. 2 shows examples of annotation for some typical images, and it can be seen that for incomplete ships, we have also effectively annotated them according to the rules.

B. Dataset Properties

As highlighted in Section III-A, the established dataset collects a large number of sea and port scene image slices. To ensure the diversity of the dataset samples, we incorporate images encompassing different platforms, ports, and both military and civilian targets. Fig. 3 shows typical examples captured under diverse detection conditions. It can be seen that targets under different conditions exhibit distinct contrast and scale characteristics. Obviously, these variations pose significant challenges to the robustness of the ship detection algorithm.

The multifaceted complexity of various environmental elements is a primary focal point during the dataset establishment process. To align algorithm training closely with real-world ship detection scenes, we have amassed images featuring diversified environmental conditions, including mist, thin cloud, thick cloud, shadows, sea clutter, and port facilities interference. It can be seen from Fig. 4 that these factors may cause image blurring, occlusion of targets, and notable false alarm disruptions. To illustrate the distribution of various scenarios in the dataset, we have provided an approximate image number of these six typical scenarios (as shown in Table III). It should be noted that in real remote sensing images, these types of environmental interference elements may couple and coexist in one image, making it difficult to divide them finely and accurately. Therefore, we provide a rough range of image numbers to provide a more intuitive and rigorous description. The diversity and richness of these environmental factors not only enhance the practical efficacy of the training algorithm but also offer valuable insights for the algorithm refinement and optimization.

Furthermore, as depicted in Fig. 5, we also performed statistical analysis on the length and width characteristics of the annotated ships in the dataset, and this analysis revealed that the ship instances exhibit a significant length–width ratio feature of approximately 5:1. To summarize, the dataset we constructed comprehensively considers the degradation

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TABLE III
IMAGE QUANTITY DISTRIBUTION FOR SIX TYPICAL SCENARIOS

Interferences	Mist	Thin cloud	Thick cloud	Shadow	Sea clutter	Port facility
Image number	≥ 600	≥ 500	≥ 300	≥ 680	≥ 960	≥ 1500

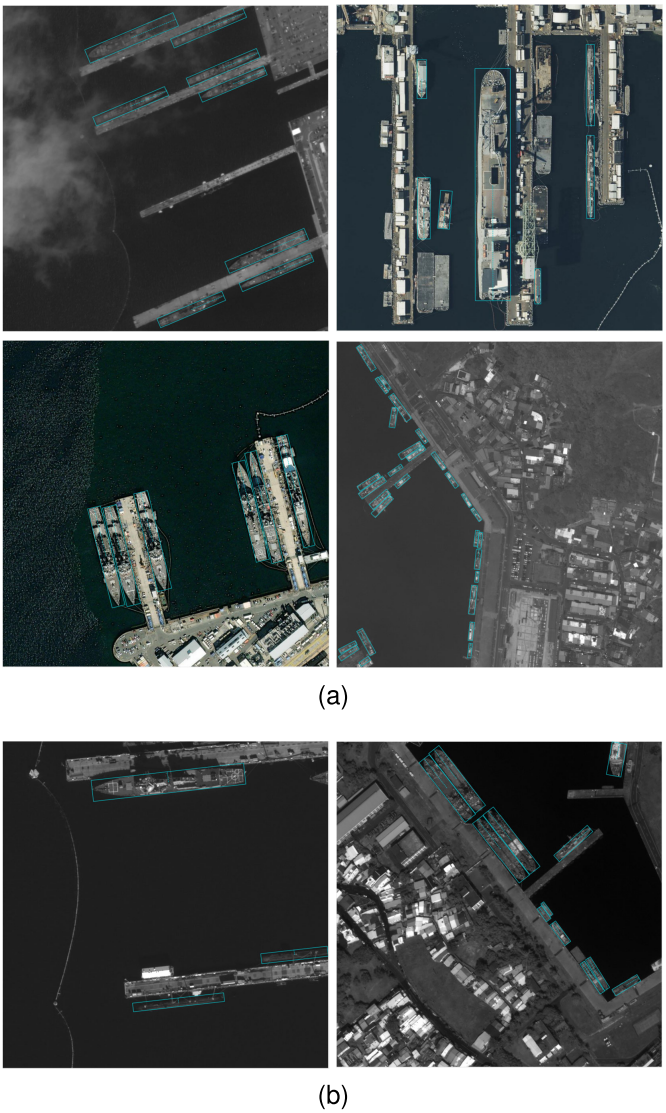


Fig. 2. Annotation results of typical images. (a) Typical scene images. (b) Images containing incomplete ships.

characteristics of different platforms, the diversity of target types, and the complexity of scene elements, and can provide basic data support for ship detection tasks in satellite images. In addition, it also brings new challenges to researchers in this field, thus driving the advancement and refinement of ship detection algorithms.

IV. SHIP DETECTION BENCHMARK

A. Evaluation Metrics

We apply the average precision (AP) and recall (R) metrics to evaluate the detection performance on the proposed dataset. The precision (P) is formulated as the prediction numbers divided by the correct prediction numbers. The recall assesses the degree of missed alarms, which is the percentage of annotation number divided by the correct prediction number. AP is calculated as the area enclosed by the curves of precision and recall, which can comprehensively evaluate the detection performance. The definitions of the above metrics are shown as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{1}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2}$$

$$\mathrm{AP} = \int_0^1 \mathrm{P}(\mathrm{R}) d(\mathrm{R}) \tag{3}$$

where TP, FP, and FN are the true positive, false negative, and false negative, respectively.

B. Implementation Details

In our experiments, to ensure fair comparisons, all the directional object detection models are implemented using the MMRotate toolbox. All experiments were conducted on a workstation equipped with an AMD EPYC 7H12 64-Core Processor (2.60 GHz CPU) and an NVIDIA GeForce RTX 3080Ti GPU. Based on the constructed dataset, we trained and evaluated 11 detection models. For a consistent comparison, we conducted training with 96 epochs and set the batch size to 4.

C. Algorithm Performance Comparison

To fully evaluate the detection performance for the proposed dataset, we select 11 classical oriented object detectors to conduct comparative experiments, including Rotated Faster R-CNN [38], RoI Transformer [30], S2A-Net [32], Redet [39], R3Det [40], Oriented R-CNN [41], SASM [42], GWD [43], Oriented RepPoints [44], G-Rep [45], and KFIoU [46]. Faster R-CNN [28], as the most influential two-stage detectors, introduces the RPN and shares the features with the backbone network, realizing the region proposal with little computation cost. RoI Transformer utilizes spatial transformations on the regions of interest (RoIs), where the transformation parameters are learned under the supervision of OBB annotations. S2A-Net employs a FAM to generate high-quality anchor points, followed by an ODM to mitigate the inconsistency between classification score and localization accuracy. ReDet introduces the rotation-equivariant network and proposes the rotation-invariant RoI Align to achieve the precision extraction of rotation-equivariant and rotation-invariant features. Oriented R-CNN applies the oriented RPN to generate the high-quality proposals with low cost. Moreover, the model employs the oriented head to refine and distinguish the RoIs. SASM proposes the shape-adaptive selection (SA-S) and shape-adaptive measurement (SA-M) strategies for sample selection and

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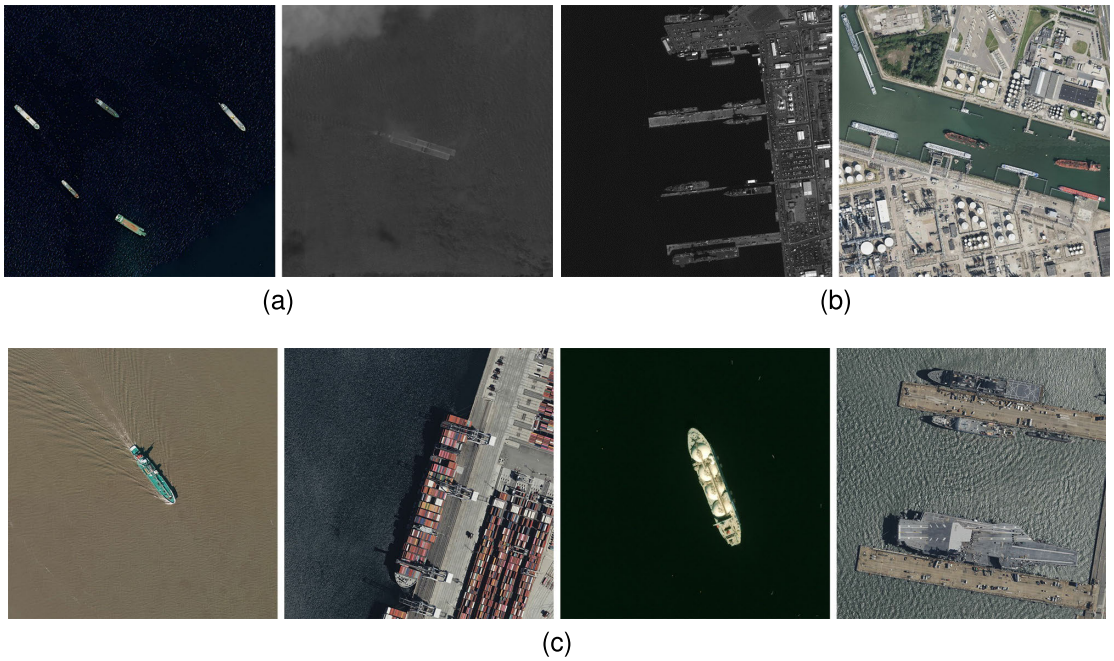


Fig. 3. Examples under different detection conditions. (a) Different platforms. (b) Different ports. (c) Different target types.

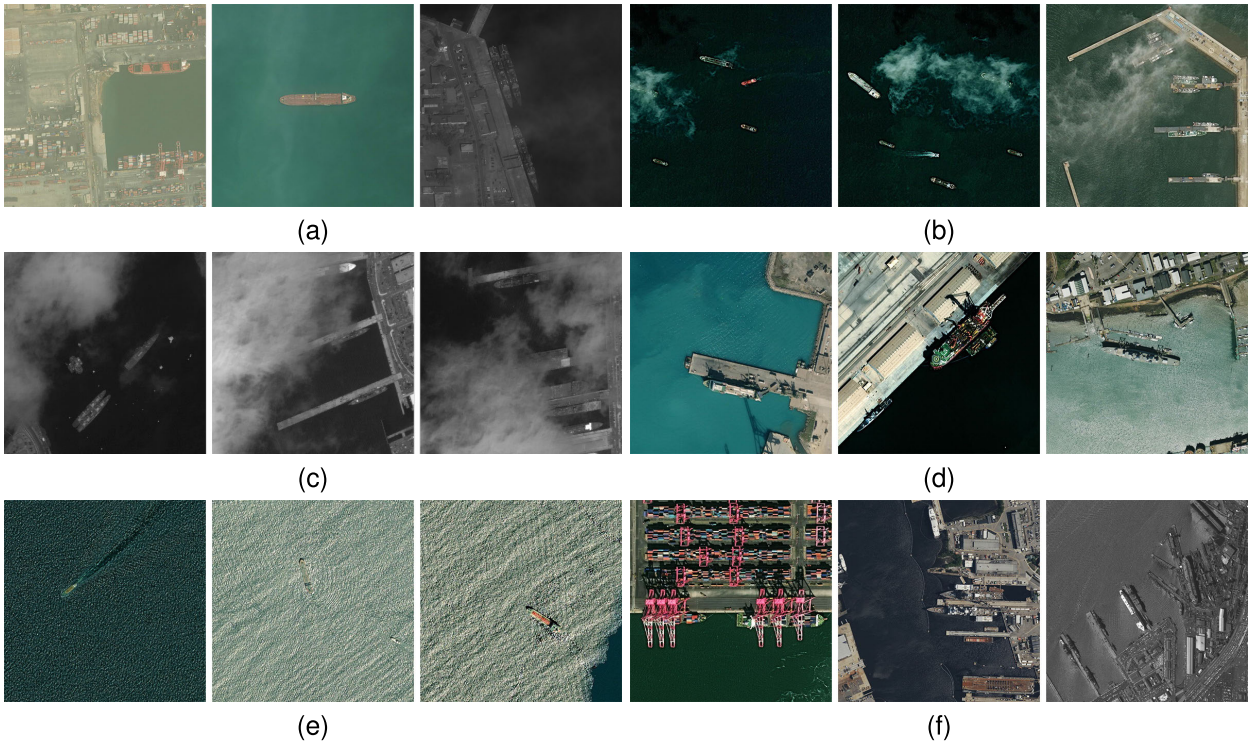
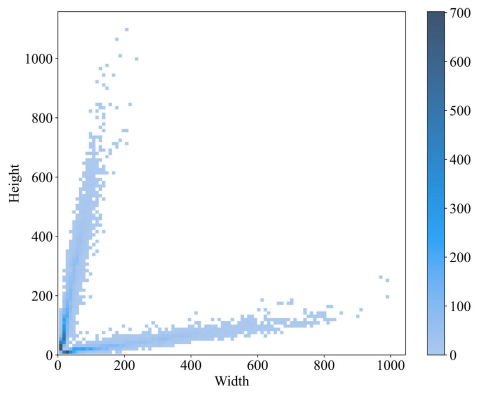


Fig. 4. Complex environmental interference situation of the dataset. (a) Mist interference. (b) Thin cloud interference. (c) Thick cloud interference. (d) Shadow interference. (e) Sea clutter interference. (f) Port facility interference.

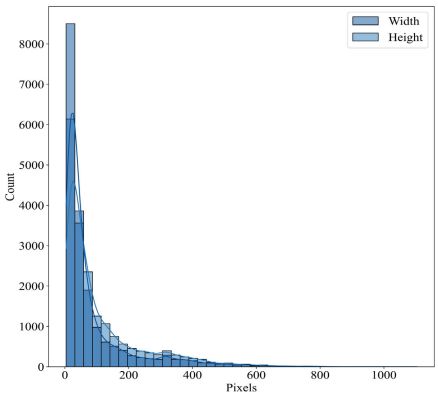
quality estimation of positive samples respectively, solving the problem of sample selection not considering target shape information and not distinguishing positive samples of different qualities. Oriented RepPoints presents an adaptive point learning strategy to extract geometric information from instances in any direction, and designs three directional transformation functions to improve the accuracy of target classification and localization.

Table IV provides the quantitative evaluation results of all considered algorithms. Overall, with the update of the algorithm release time, the detection performance on this dataset basically shows a positive trend of improvement. Specifically, Rotated Faster R-CNN is proposed a long time ago and performs the worst, with 0.620 AP and 0.745 recall. In comparison, KFIoU proposed in 2023 obtains the most promising performance with 0.835 AP and 0.906 recall, which

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(a)



(b)

Fig. 5. Scale distribution of the targets in the dataset. (a) Height and width distribution. (b) Area distribution.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT ALGORITHMS

Algorithm names	Release year	Backbone	Epoch	AP	Recall
Rotated Faster R-CNN	2017	ResNet50	96	0.620	0.745
RoI Transformer	2019	ResNet50	96	0.745	0.836
S2A-Net	2021	ResNet50	96	0.722	0.817
Redet	2021	ReResNet50	96	0.798	0.863
R3Det	2021	ResNet50	96	0.781	0.846
Oriented R-CNN	2021	ResNet50	96	0.784	0.844
SASM	2022	ResNet50	96	0.743	0.892
GWD	2022	ResNet50	96	0.792	0.867
Oriented RepPoints	2022	ResNet50	96	0.823	0.902
G-Rep	2023	ResNet50	96	0.818	0.894
KFIoU	2023	ResNet50	96	0.835	0.906

is 0.12 higher than the oriented RepPoints algorithm with the second highest AP value. Moreover, S2A-Net achieves the second worst performance, with the AP of 0.722, which indicates that one-stage detectors are inferior to two-stage detectors generally when applied to ship detection in complex scenes. In 2023 algorithm comparison, Redet and oriented RepPoints show an advantage in terms of AP value, respectively.

To more intuitively demonstrate the application effects of various algorithms on the dataset, Fig. 6 visualizes the annotations and detection results of Rotated Faster R-CNN, SASM, and oriented RepPoints methods. We have selected six representative scenes, corresponding to thin clouds, sea clutter under complex lighting, shadows, multiscale targets, thick clouds, and complex port scenes. For clear display, we utilize green boxes to illustrate the annotation of ground truth. In the detection results, red boxes represent correctly detected targets, yellow boxes represent false alarms, and missed detections are not displayed.

TABLE V
DETECTION RESULTS OF THE ROTATED FASTER-RCNN METHOD

Scenes	Epoch	Recall	AP
Mist	36	0.791	0.786
Thin cloud	36	0.835	0.814
Thick cloud	36	0.737	0.711
Shadow	36	0.788	0.787
Sea clutter	36	0.787	0.781
Port facilities	36	0.745	0.752

From the detection results, it is evident that for various complex environmental interferences, the selected methods can detect ship targets of different scales and directions to varying extents. This effectiveness is mainly due to the advantage of deep networks being able to learn target features at multiple levels. However, Rotated Faster R-CNN suffers from inaccurate angle regression, missed detections, and high false alarm rates in these complex scenes, which indicates that deep networks are difficult to accurately and comprehensively characterize the characteristics of targets using only conventional RPN structures. Although SASM can more effectively suppress background interference compared to the former method, there are still some missed detections and false alarms when detecting small targets. This issue arises because this method focuses on adaptive learning and measurement of target shape information, but the small-scale target lacks shape information, and remote sensing images may have a degree of blurriness due to the degradation effect of transmission links, thus limiting the performance of this method. Oriented RepPoints has the highest AP value among these three models. Although there are still a few false alarms and missed detections against sea clutter and cloud interference, its visualization results also demonstrate the relatively good robustness of this method in complex scenes. Consequently, the complex environmental interference factors of the dataset still pose many challenges to typical advanced methods, highlighting that our dataset can spur the refinement and advancement of existing algorithms.

D. Additional Experiment

In order to analyze the impact of six typical scenarios in the dataset on ship detection performance, we mainly apply two representative methods, Rotated Faster R-CNN and Oriented Reppoints, to test six typical scenarios separately. The detection results of the two methods are shown in Tables V and VI.

From the tables, it can be found that both methods perform the worst in thick cloud scenes in term of AP and recall, mainly because thick clouds can cause significant occlusion of ship targets, resulting in missing target features and subsequent

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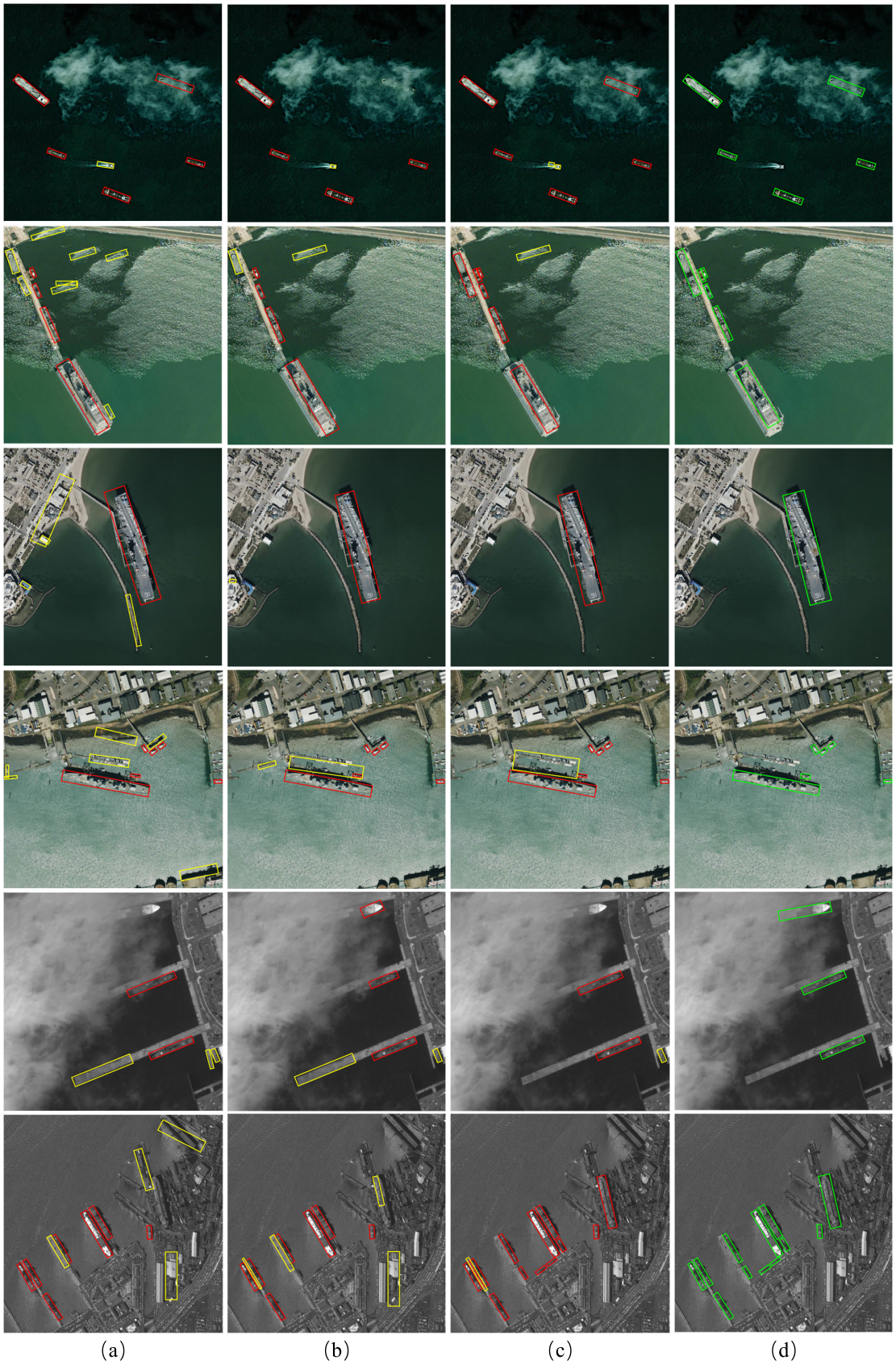


Fig. 6. Detection results provided by different baseline models. (a) Rotated faster R-CNN, (b) SASM, (c) oriented RepPoints, and (d) ground truth.

reducing detection performance. The second worst performance of the two methods in port scenarios is mainly due to the high similarity in shape, color, and other characteristics between artificial port facilities and ship targets, which can

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TABLE VI
DETECTION RESULTS OF THE ORIENTED REPPOINTS METHOD

Scenes	Epoch	Recall	AP
Mist	36	0.834	0.862
Thin cloud	36	0.860	0.889
Thick cloud	36	0.796	0.806
Shadow	36	0.863	0.873
Sea clutter	36	0.839	0.859
Port facilities	36	0.808	0.824

easily lead to false alarms and missed detections. Relatively speaking, mist, sea clutter, and shadows have similar effects on target detection, reflected in relatively small fluctuations in evaluation indicators. It is worth noting that the two methods perform the best in detecting thin cloud scenes, mainly because under the condition of image resolution in the dataset, the ship size is relatively large, and thin clouds may have a certain blurring effect on the cloud layer, but they can still effectively preserve the shape and texture features of the ship targets. The above results can not only prove that different scenarios have different impacts on ship detection, but also reveal the severity of the impact of different scene elements on the performance of ship detection methods. This achievement can provide a new perspective for the research of ship detection methods.

V. CONCLUSION

In this article, we construct a ship detection dataset titled SCCOS comprising 4639 images with 19 078 instances. To ensure data diversity, this dataset is aggregated from a variety of platform sources, including Google Earth, Microsoft Map, Worldview-3, Pleiades, Ikonos, Orbview-3, and Jilin-1 satellites. Moreover, to make the dataset closely mirrors complex scenes encountered in real-world applications, we have comprehensively considered complex environmental elements such as thin clouds, mist, thick clouds, light shadows, sea clutter, and port facilities.

Additionally, we have conducted evaluation experiments using 11 advanced detection algorithms on this dataset, establishing a performance benchmark. The experimental results demonstrate that the proposed dataset presents substantial challenges, particularly regarding the multiscale characteristics of targets, interference from environmental factors such as thin and thick clouds, and interference from similar objects in port areas. Existing representative methods tend to exhibit issues with false alarms and missed detections. Future efforts need to integrate the optical characteristics of ship targets within images for subsequent refinement and optimization. Therefore, the established dataset and performance benchmark of the dataset can effectively facilitate research in intelligent processing and analysis of remote sensing images and their associated applications.

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